

FCU3

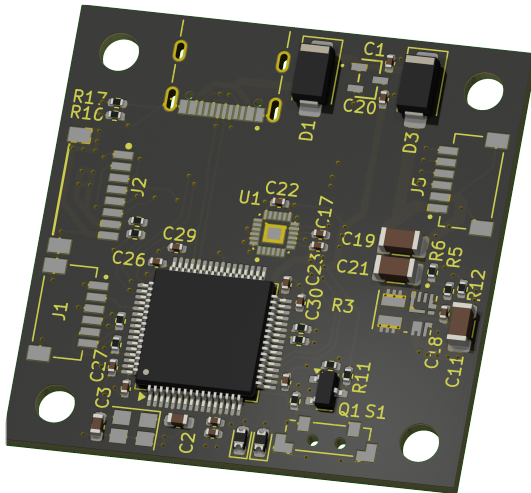
Variant: CHECKED

2025-12-13

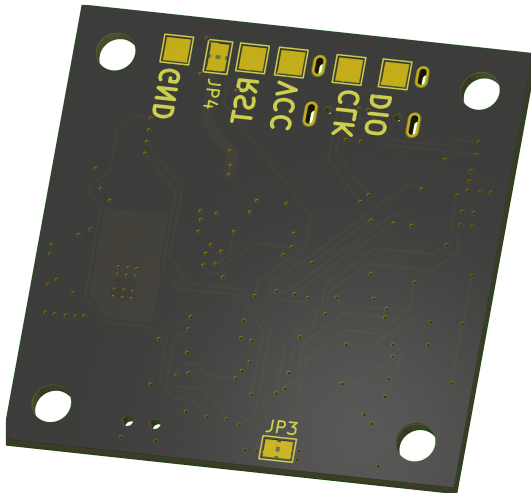
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 13-Dec-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

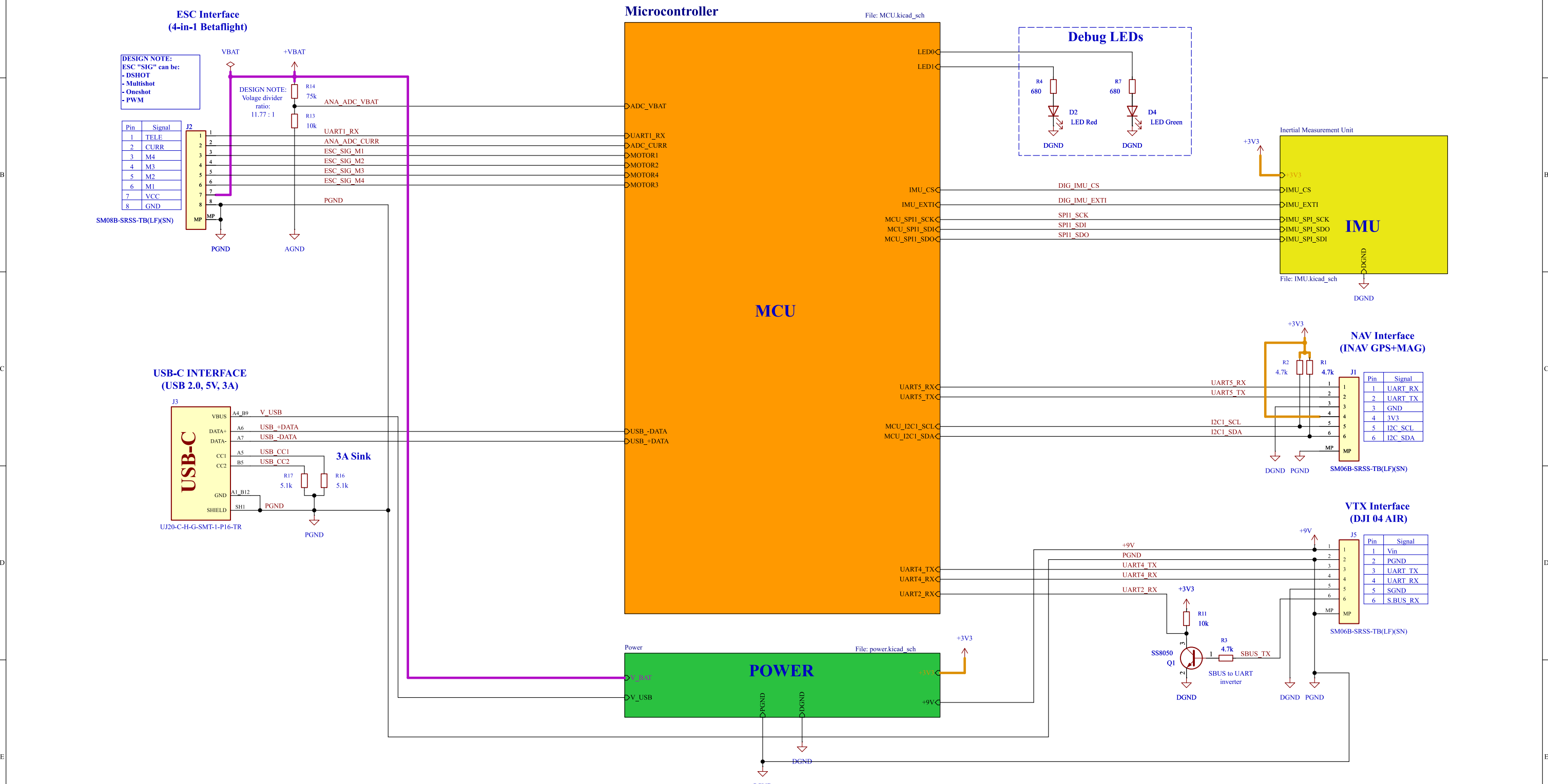
Example text for critical layout guidelines.

	Comments:	Company: Kaldstrand Electronics	Variant: CHECKED	Git Hash: 4dad50
	Sheet Title:	Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Path: /	File Name: FCU3.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12
			Reviewer:	Revision: + (Unreleased)
			Size: A3	Sheet: 1 of 8

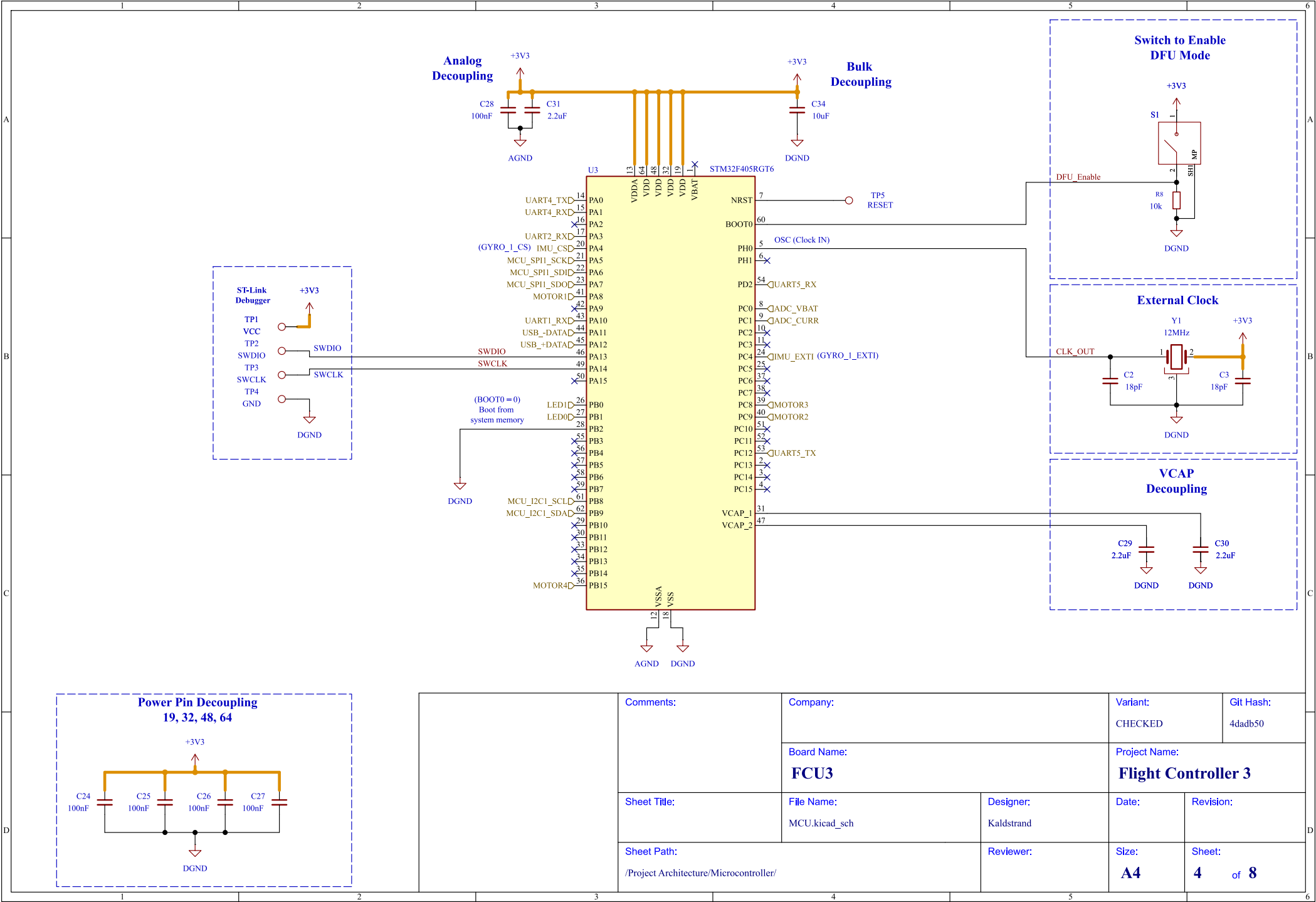
[illegible]

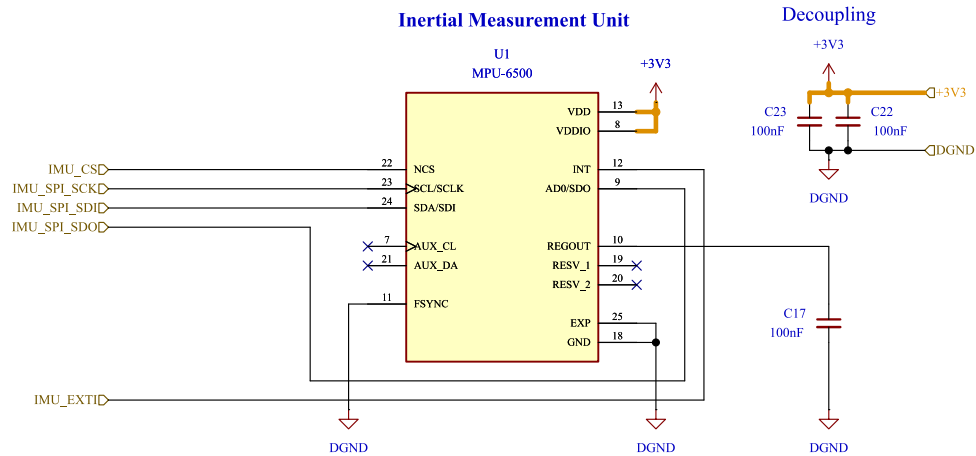
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

[3] Project Architecture

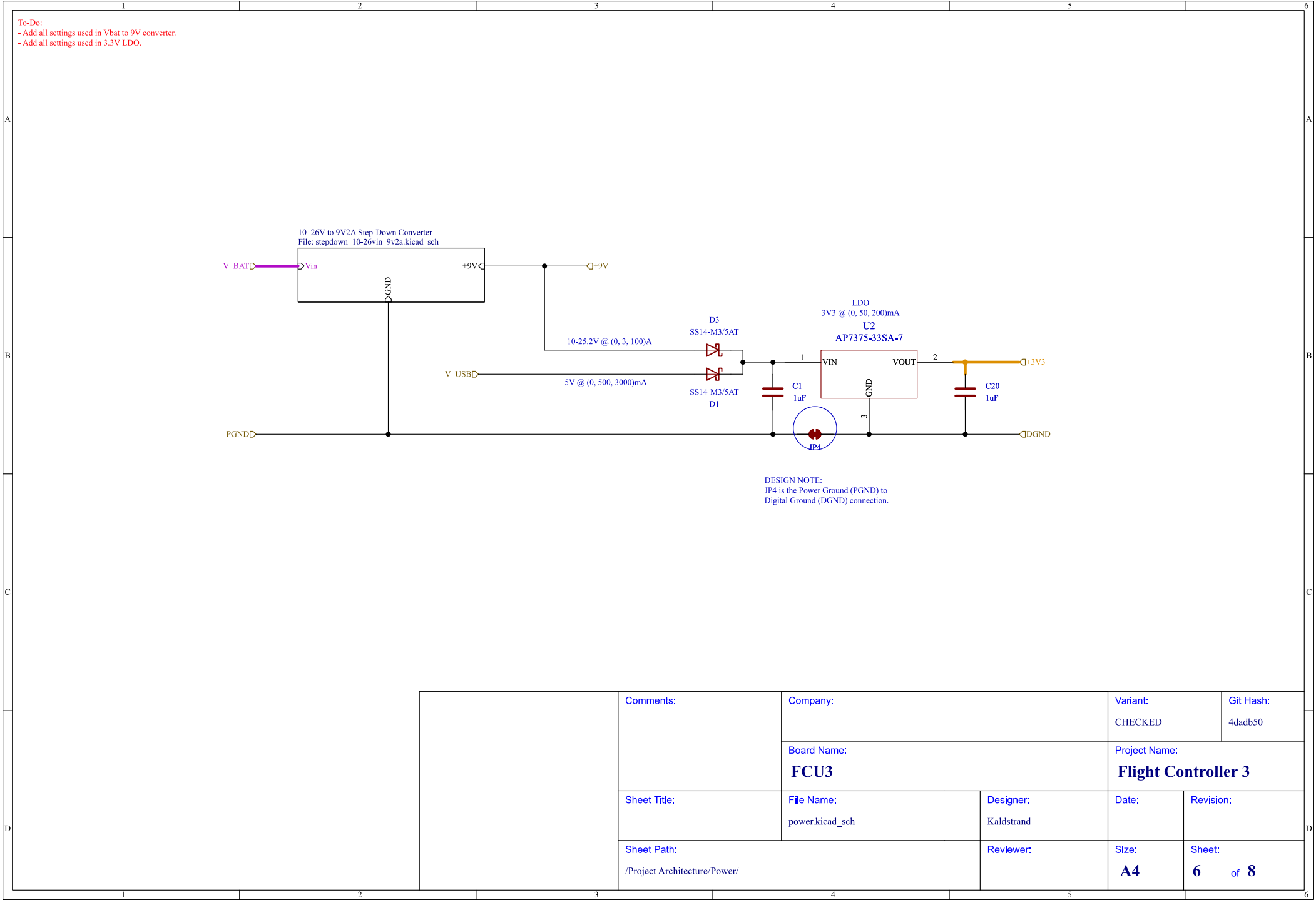


Grounding Strategy		Comments:		Company:		Variant:	Git Hash:
		Project Architecture		Kaldstrand Electronics		CHECKED	4dadb50
Sheet Title:		File Name:		Board Name:		Project Name:	
		Project Architecture.kicad_sch		FCU3		Flight Controller 3	
Sheet Path:		Designer:		Date:		Revision:	
		Kaldstrand		2025-01-12		+ (Unreleased)	
/Project Architecture/		Reviewer:		Size:		Sheet:	
				A3		3 of 8	





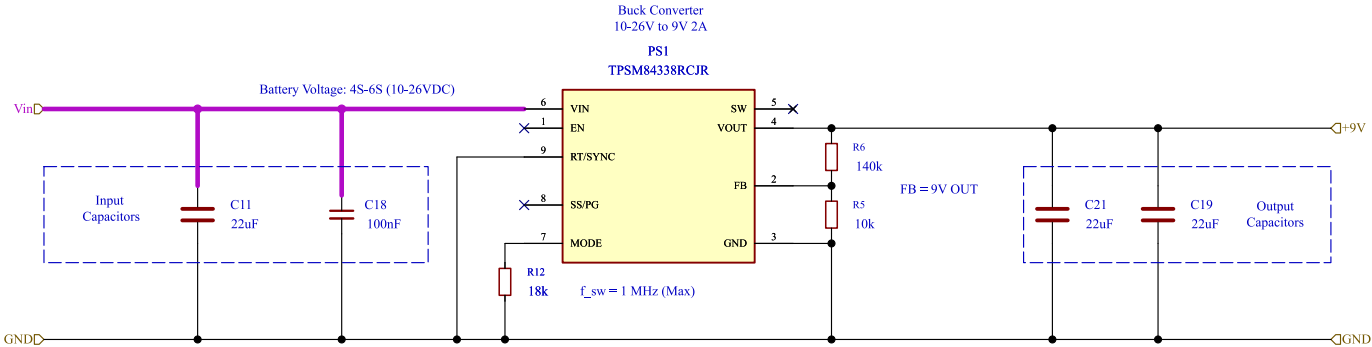
	Comments:	Company:	Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4 Sheet: 5 of 8



DESIGN NOTE:
 $R_{FBB} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBB} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBB} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R11 = 10k, R6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:	Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: stepdown_10-26vin_9v2a.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Power/10-26V to 9V2A Step-Down Converter/	Reviewer:	Size: A4	Sheet: 8 of 8

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8